Def. If $s: X \to [0,\infty)$ is a measurable simple function with representation

$$s = \sum_{i=1}^{n} a_i \cdot \chi_{A_i} \quad \text{where} \quad \text{range}(s) = \{a_1, \cdots, a_n\}.$$

For each $E \in \mathcal{M}$, define

$$\int_E s \, d\mu \overset{\text{def}}{=} \sum_{i=1}^{n} a_i \cdot \mu(A_i \cap E).$$

For a measurable function $f: X \to [0,\infty]$, and $E \in \mathcal{M}$, define

$$\int_E f \, d\mu \overset{\text{def}}{=} \sup \left\{ \int_E s \, d\mu \;\middle|\; s \text{ is a simple function s.t } 0 \le s \le f \right\}.$$

This integral is called a Lebesgue integral of f over E, with respect to the measure μ.